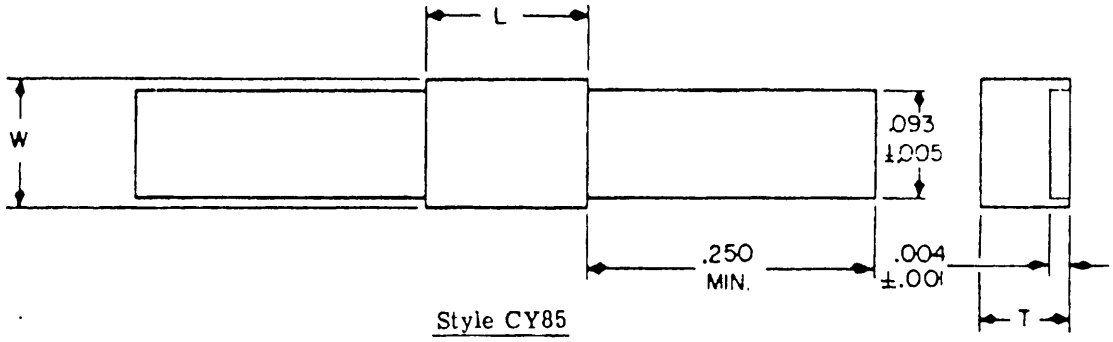
MILITARY SPECIFICATION SHEET

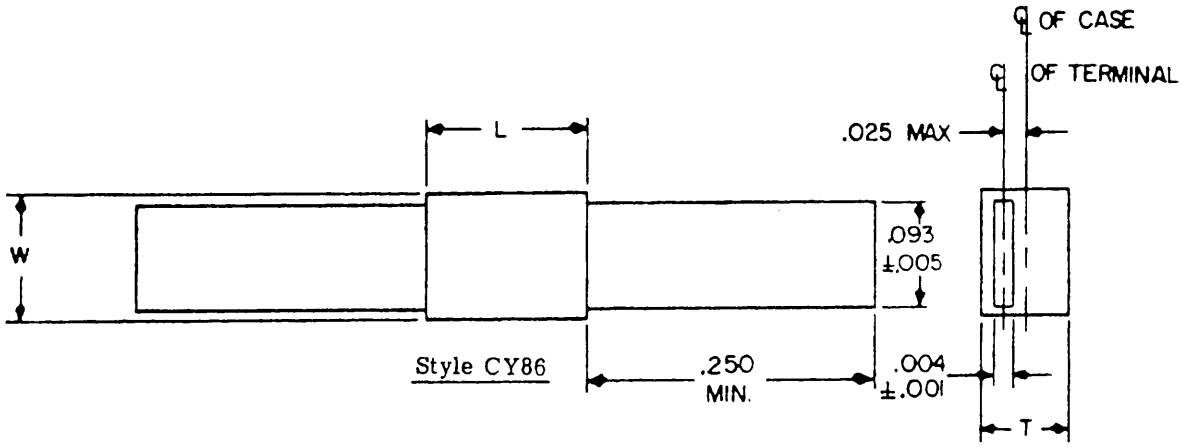
CAPACITORS, FIXED, GLASS DIELECTRIC, STYLES CY85, CY86, CY87, CY88, AND CY89

This specification is approved for use by all Departments and Agencies of the Department of Defense.

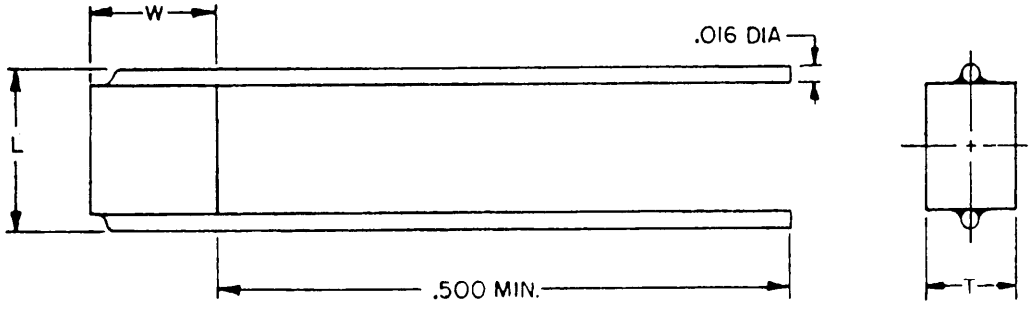
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11272.



Style CY85



Style CY86

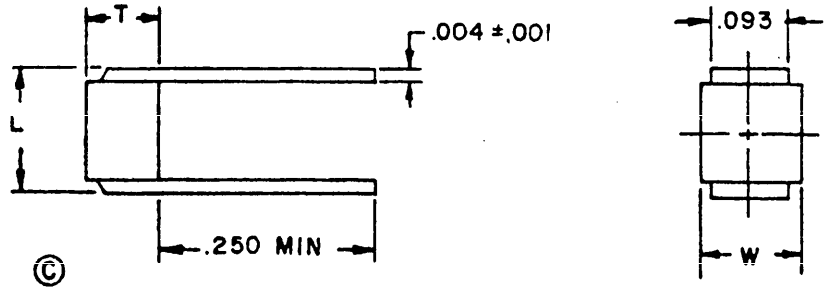


Style CY87

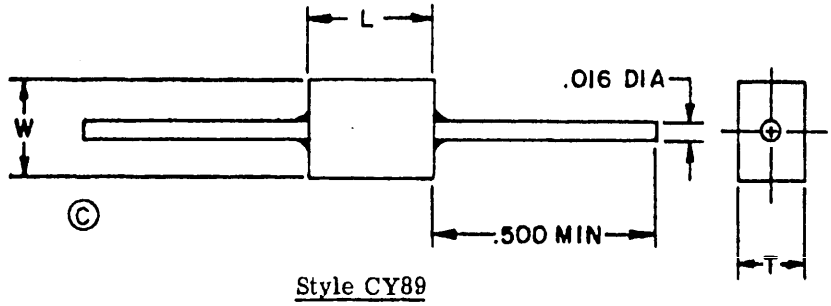
© denotes changes

FSC 5910

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Style CY88



Style CY89

INCHES	MM	INCHES	MM
.001	.03	.093	2.36
.004	.10	.100	2.54
.005	.13	.110	2.79
.015	.38	.135	3.43
.016	.41	.250	6.35
.025	.64	.500	12.70
.060	1.52		

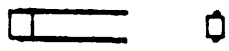
© Dimensions				
Style	L	W	T	
			Min	Max
CY85	.135 ±.015	.110 ±.015	.060	.100
CY86	.135 ±.015	.110 ±.015	.060	.100
CY87	.135 ±.015	.110 ±.015	.060	.100
CY88	.135 ±.015	.110 ±.015	.060	.100
CY89	.135 ±.015	.110 ±.015	.060	.100



Style CY85



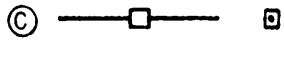
Style CY86



Style CY87



Style CY88



Style CY89

APPROX. SIZE

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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TABLE I.

Type designation 1/	Capacitance	Rated voltage	Capacitance tolerance available
	pF	Volts, dc	
CY8-D0R1B	0.1	500	B
CY8-D0R2B	0.2		B
CY8-D0R3-	0.3		B, C
CY8-D0R4-	0.4		B, C
CY8-D0R5-	0.5		B, C, D
CY8-D0R6-	0.6		
CY8-D0R7-	0.7		
CY8-D0R8-	0.8		
CY8-D0R9-	0.9		
CY8-D1R0-	1.0		
CY8-D1R1-	1.1		
CY8-D1R2-	1.2		
CY8-D1R3-	1.3		
CY8-D1R4-	1.4		
CY8-D1R5-	1.5		
CY8-D1R6-	1.6		
CY8-D1R7-	1.7		
CY8-D1R8-	1.8		
CY8-D1R9-	1.9		
CY8-D2R0-	2.0		
CY8-D2R1-	2.1		
CY8-D2R2-	2.2		
CY8-D2R4-	2.4		
CY8-D2R7-	2.7		
CY8-D3R0-	3.0		
CY8-D3R3-	3.3		
CY8-D3R6-	3.6		
CY8-D3R9-	3.9		
CY8-D4R3-	4.3		
CY8-D4R7-	4.7		
CY8-D5R1-	5.1		
CY8-D5R6-	5.6		
CY8-D6R2-	6.2		

See footnote at end of table.

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TABLE I. - Continued

Type designation 1/	Capacitance	Rated voltage	Capacitance tolerance available
	pF	Volts, dc	
CY8-D6R8-	6.8	500	B, C, J, K, M
CY8-D7R5-	7.5		
CY8-D8R2-	8.2		
CY8-D9R1-	9.1		
CY8-D100-	10		F, G, J, K, M
CY8-D110-	11		
CY8-D120-	12		
CY8-D130-	13		
CY8-D150-	15		
CY8-D160-	16		
CY8-D180-	18		
CY8-D200-	20		
CY8-D220-	22		
CY8-D240-	24		
CY8-D270-	27		
CY8-D300-	30		
CY8-D330-	33		
CY8-D360-	36		
CY8-D390-	39		
CY8-D430-	43		
CY8-D470-	47		
CY8-D510-	51		
CY8-D560-	56		
CY8-D620-	62		
CY8-D680-	68		
CY8-D750-	75		
CY8-D820-	82		
CY8-D910-	91		
CY8-D101-	100		
CY8-D111-	110	300	
CY8-D121-	120		
CY8-D131-	130		
CY8-D151-	150		
CY8-D161-	160		
CY8-D181-	180		
CY8-D201-	200		
CY8-D221-	220	200	
CY8-D241-	240		
CY8-D271-	270		
CY8-D301-	300		
CY8-D331-	330		
CY8-D361-	360		
CY8-D391-	390		
CY8-D431-	430		
CY8-D471-	470		
CY8-D511-	510	100	
CY8-D561-	560		
CY8-D621-	620		
CY8-D681-	680	50	
CY8-D751-	750		
CY8-D821-	820		
CY8-D911-	910		
CY8-D102-	1000		

1/ Complete type designation will include an additional digit to indicate the style, and an additional letter symbol to indicate the capacitance tolerance, where applicable.

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REQUIREMENTS

Design and construction:
Dimensions and configuration: See figure and table I.
Case type: Multi-layer, unencapsulated, monolithic.
Material - Porcelain.
Terminals: Styles CY85 and CY86 - Silver leads attached with solder, if used, with melting temperature less than 200°C. Style CY87-26 gauge wire, solderable as to MIL-STD-202E, method 208C.
Inspection conditions: All visual examinations shall be performed using a stereo microscope having a minimum magnification of 20 power.
Capacitance (Cap.) value: See table I.
Cap. tolerance: See table I.
Rated voltage: See table I, characteristic D.
Operating temperature range: -55°C to +125°C.

Insulation resistance (IR):
At +25°C: 10^6 megohms, minimum (for 0.1 pF thru 470 pF).
10^5 megohms, minimum (for 510 pF thru 1,000 pF).
At +125°C: 10^5 megohms, minimum (for 0.1 pF thru 470 pF).
10^4 megohms, minimum (for 510 pF thru 1,000 pF).

Quality factor: 10,000. 1 MHz -0 +100 kHz.

Life:
At +25°C and +125°C:
IR - Not less than initial requirement.
Cap. - Change not more than 0.2 percent or 0.2 pF, whichever is greater, from the nominal value.

Temperature coefficient and capacitance drift. See table II.

TABLE II. Temperature coefficient and capacitance drift.

Temperature coefficient	Capacitance drift (-55° to +25°C)
Parts/million/°C 90 ±20	0.1 percent or 0.1 pF, whichever is greater

NOTE: These capacitors are intended for use in RF solid-state circuitry up to and including 25 gigahertz, and cannot be replaced in the field.

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-F238)

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